The Law of Graphic Equalisation — Mind and Self Grounding

Frames, the Run and the Record, Free Will, Self-Models, Personal Identity, and Death

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DRAFT — draft-2026-08-27 (27 August 2026). This document is under active revision and is circulated for comment, not as a finished result. It is a companion to *The Law of Graphic Equalisation — Theory* (referred to below as Graphic Equalisation) and to its physics companion *The Law of Graphic Equalisation — Physics Grounding* (the physics companion), and inherits their definitions, epistemic boundary, and level-of-claims. Sections marked as falsifiability commitments are the intended points of attack.

Abstract

This companion to Graphic Equalisation develops the mind-and-self groundings of the change-substrate. Its central move is that a frame is real graph data that *runs*, and that everything here follows from the run and its record. A rock and a universe are both frames, so anything a rock does without is not compulsory for a frame. Recording spatialises a succession into a static held axis, giving the run/record axis whose poles are Conway (pure running, computationally irreducible) and a movie (pure record), with real frames the machinima between; the past can be a record but the future is uncomputed, so it is open as *unrun*, not as indeterministic. Free will is then the in-frame run — a fully recorded, zero-randomness story demonstrates it, its characters free in frame while the reader foreknows — so freedom needs no indeterminacy. Imagination is real: there is no non-actual frame, and the only difference between the imagined and the perceived is frame-relative grounding. Self-models are incomplete by construction: a frame carries its own trajectory rather than reading it, so to treat that course as an object at all it must build a model of itself, under a finite aperture. The incompleteness is structural rather than pathological, and it divides by kind — a limit of *expression* leaves residual and is inferable from trace, while a limit of *aperture* leaves nothing and is inferable only by metacognition — so the operative distinction is calibrated versus uncalibrated, not complete versus incomplete. Death is the self's rules ceasing to be re-runnable in the same coherent self; personal identity is internal coherence (with an instance a supergraph node), which places gradual replacement, teleporter, backup, and fission on one axis; and death is relative to a chosen coherence-boundary, with many real deaths and none absolute. The claims throughout are information-level and structural.

1 Scope

The method is to treat philosophy of mind as applied physics: once a frame is graph data that runs, free will, identity, and death are read off the run and its record, not argued as separate doctrines. This paper is a grounding, not a psychology and not a metaphysics asserted from within. It inherits Graphic Equalisation's position that the claims are about information — distinction, frame, admissibility, realisation, and now the *run* of a frame and its *record* — of which mind is one embodiment. The strong claims are structural: that free will is the in-frame run and needs no indeterminacy; that imagination is real and differs from perception only in

grounding; that every self-model is incomplete by construction, so the operative distinction is calibrated versus uncalibrated rather than complete versus incomplete; that personal identity is internal coherence (Graphic Equalisation's SELF identity), distinct from the supergraph node that instances it (its SUPERGRAPH identity); and that death is the loss of re-runnability in the same coherent self, relative to a chosen coherence-boundary. The weak part is anything operational: no decision procedure for a given choice, no test for when re-runnability or coherence has in fact been lost, no threshold for how gradual a change stays "the same" self, and no account of the subjective, felt, or phenomenal character of any of it — consciousness and qualia are out of scope. Each grounding is a *promoted* constraint exposed to falsification, in the manner of Graphic Equalisation's collapse bridge, not a result asserted from within.

2 The minimal universe

A *rock*, in the sense used throughout, is an inert frame: a graph whose rules are present but not firing, so it has no succession and no time of its own. Things happen to it from its supergraph, but in its own frame it merely is (the physics companion develops this as why a rock has no time). Everything here is a graph, so a rock and a universe are one kind of thing, differing in scale, not in category (Graphic Equalisation's self-similarity: the runner is graph-that-runs at every scale). A rock *is* a universe — the minimal one — and that makes it an existence proof about universes in general.

Proposition 2.1 (What the rock shows is not compulsory)**.** *Anything not compulsory for a rock is not compulsory for a frame, and the universe is one frame among others. The rock does without time (it is inert, no succession), without cycles or closure, without inference or observation, without history or records, without self-originated change, and without a running arrow — so none of these is compulsory for a universe.*

The inference is one-directional, and only that direction is claimed. A single inert example bounds the compulsory set from *above* — it shows what can be dispensed with — and cannot establish that what the rock happens to retain is necessary. So the rock is a non-necessity argument, not a derivation of a core.

What is compulsory is inherited from Graphic Equalisation, which keeps almost nothing: a container relation across which rules run. The rock is consistent with that and does not weaken it, because a rock's rules are inert *in its own frame* while its container's are not — what happens to the rock (weathering, being moved) is its supergraph's rules running across exactly that container relation. An inert frame is therefore one whose own rules do not fire, not a counterexample to rules running.

Nor does that demotion cost the argument anything, because *frame* is the category throughout. The universe is a frame and a rock is a frame; they differ in scale, not in kind, which is what self-similarity asserts. So the inference never needed "universe" as a separate qualification — it runs over frames, and reads: anything a frame can be without is not compulsory for a frame. The rock's inertness is not a defect in the example but the whole of its content.

This is the general form of "time is optional," and it licenses the contingent-not-necessary verdicts of the physics companion. Wherever a shape there was called contingent rule-content rather than necessity — the number of spatial dimensions, the gauge group, the dimensionless constants, the arrow, the value of c — the rock is the reason: the minimal universe does without it, so it cannot be compulsory. Physics ordinarily takes time, causality, and the second law's arrow as features a universe must have; the rock shows they are features an *active* universe has.

3 Running and recording: Conway, a movie, and the machinima between

Recording spatialises time. A succession held as a record — an MRI of a rock, a film strip, a worldline drawn on paper — lays the temporal order out as a static structure, a +-like axis (the block's extra dimension). This reads the metric signature: the − axis is the one being *lived*, the frame's own succession run by its cycles, open at the frontier; the + axes are *held* — spatial positions, and any recorded time. The sign tracks lived-versus-held, not a property intrinsic to the axis; the block feels dead because a recorded succession is not run.

The extremes of the run-versus-record axis are Conway and a movie. Conway's Game of Life is pure running: a rule and a seed, everything computed live, nothing recorded, and no way to know the outcome but to run it (computational irreducibility). A movie is pure record: every frame laid out, played back, nothing computed — the block universe, known by reading rather than running. Between them is scripted machinima — a running engine following a predetermined script, part computed live, part laid down in advance — and that is where real frames sit: rules fire (Conway-like) while history is held (movie-like). It is the same axis as generation-versus-record: store the generator and run it (Conway, small but irreducible) or store the frames and read them (a movie, large but immediate), with the world's own compressibility deciding which is cheaper.

Proposition 3.1 (The past can be a movie; the future is Conway)**.** *A block is only ever of the past. What has been run can be recorded and laid out — a movie of the succession so far, readable without re-running. What has not been run cannot be recorded, because there is nothing yet to record: the future is determinable by the rules but* uncomputed*, Conway-style, so it is not laid out and cannot be read ahead — you must run it. The open frontier of Axiom* −1 *is therefore not indeterminism but unrun-ness: the future is open because it is uncomputed, not because nothing fixes it, and you cannot record what has not run.*

4 Free will and the three frames

Proposition 4.1 (Free will is the in-frame run; a story demonstrates it)**.** *Free will is a frame running its own decision. In a frame's own run the decision is live and unrun-to-it — the frame must run its deliberation to reach the outcome — and that running* is *the choosing: the outcome is fixed by the rules yet knowable only by running, so the frame experiences computing it as choosing it. This needs no indeterminacy, and a story is the perfect demonstration. A book or a movie is a fully recorded succession — every word or frame fixed, zero randomness, readable ahead by the reader — and yet its characters have complete free will: they deliberate, choose, and are responsible, in frame. Both hold at once because they are two frames. The character's freedom is its in-frame run; the reader's foreknowledge is a superframe holding the finished record (a movie), and holding the record never reaches into the character's run to remove the freedom. So foreknowledge is not unfreedom — not even complete foreknowledge, the reader who knows the ending or the author who wrote it — because knowing the recorded run and being the live run are different frames. Randomness would only subtract: a random run is not more free, it is less the character's own. And responsibility survives for the reason the freedom does — the run is the character's own (self-change is the only change; the rules running are the character) — so a determined, recorded, even foreknown act is still that character's act.*

Remark (Author, character, reader: unrun-ness is absolute)*.* A story stacks three frames on one succession. The author *runs* it — writing is Conway, generated live, and the author must write to know the ending (authorship is computationally irreducible; the commonplace that "the characters took on lives of their own" is the author unable to read ahead their own creation). The character *lives* it — the in-frame run of Proposition 4.1. The reader *replays* it — the

finished record, a movie. One succession, three frame-readings: the run/record axis is a fact about frame, not about the system. And the author settles the objection to Proposition 3.1 that a high enough superframe might read the future ahead — even the creator gets no shortcut. The author is Conway to their own book while writing it; the movie exists only *after* the run, for everyone including the author. Foreknowledge is always post-run record-holding, never pre-run read-ahead. For an irreducible rule unrun-ness is absolute: no frame, creator included, reads an unrun future without running it; every frame either runs it or replays an already-run record.

Remark (The reader runs and holds at once, and this is standard). The reader looks like a counterexample: the account needs the character's frame genuinely to run, and needs the reader to hold a finished record — and if the character's frame runs *in* the reader, the two frames appear to collapse into one. They do not, and the point does not need arguing here, because three separate literatures settled it independently.

The cleanest statement is Currie's: a reader of fiction simulates the state of a hypothetical *reader of fact* (Currie, 1997; Currie and Ravenscroft, 2002). Running and record-holding are then not competitors but the same fact, since what is run is a simulation and what it simulates is someone holding a record. Rhetorical narratology reaches the same place by dividing the reader rather than the frame: Rabinowitz (1977) distinguishes the *narrative* audience, for whom the narrated events are real, from the *authorial* audience, which knows the text is made, and these are positions one actual reader occupies simultaneously (Polvinen and Sklar, 2019). The two frames were never reader-versus-character. And the cognitive architecture supplies the mechanism: pretence representations are processed by the same inference machinery that operates over beliefs — a single code (Nichols and Stich, 2000) — while *quarantining* confines their effects to a circumscribed domain (Gendler, 2003). Same machinery is not same frame. Walton (1978) states the resulting two-level structure exactly: what runs in the appreciator is real (his "quasi-fear" — the tensed muscles, the quickened pulse) and it is that real state which makes it make-believe that he is afraid, with full knowledge that the slime is not. Literary theory has a name for the compound, *aesthetic illusion*: immersion counterbalanced by a latent rational distance, the awareness of the difference between representation and reality (Wolf, 2014).

Nor is it only characters that are modelled. A reader builds a model of the *author* as well, and that construct is long established: Booth's *implied author*, the version of the writer a reader inevitably assembles from the text (Booth, 1983), which the standard treatment takes to have both an objective side, grounded in textual indexes, and a subjective one, since those indexes are weighed differently by each reader (Schmid, 2014). Genette (1988) puts it most economically — the implied author is everything the text lets us know about the author — and hypothetical intentionalism makes the reasoning explicit, locating a work's meaning in what an appropriate audience would most reasonably hypothesise the situated author to have meant by composing this text (Levinson, 2002). That readers infer a mind behind a work at all has empirical support: viewers distinguish professional abstract works from superficially similar ones by children and animals even when the labels are switched, and justify the professional choices by reference to the artist's intentions (Hawley-Dolan and Winner, 2011).

The mode is the one this paper has already named. A frame cannot read its author's frame; it has only what that frame produced, and so infers it — the +1 direction of the level stack, run on a person rather than a container. Sustained output makes that inference rich, since a generator sampled at length discloses a great deal of itself whether or not it means to.

That such a model is *generative* — runnable forward on inputs the author never addressed — is the part the prior art does not supply, and it is worth being exact about its status, because it is not a new posit. It is a composition of three things each established independently. An author-model is a model. Models can approach the behaviour they model arbitrarily closely, given enough of it to work from. And analysis by synthesis runs in both directions: a recogniser built that way recognises *by* generating candidates and comparing them, so it contains the generator as a matter of construction rather than as an addition to it. That architecture is

not speculative. It is the modern form of the predictive account already drawn on above — perception as the inversion of a generative model, predicted input compared against actual (Friston, 2010) — and it has been tested directly on people: three-dimensional shape perception is best explained as analysis by synthesis, matching human observers in accuracy and response time better than the alternatives, and alone correlating with human performance on the hard discriminations (Yildirim et al., 2024). A generative author-model follows from those, and needs no further machinery.

What that argument does not establish is that any particular reader's model is of this kind, and the distinction matters because the obvious evidence does not decide it. Recognising a style requires no model of its author at all: pigeons trained on Monet and Picasso discriminate the two and generalise to unseen paintings (Watanabe et al., 1995). Discriminative recognition is available without any generator, so neither recognising a voice nor sensing that a pastiche is wrong shows anything about which kind of model is in play. What would is the forward direction — what the author would have made of something they never saw, right or wrong in ways others can check — and how far a given reader can go there is, like the rest of this, a position on several axes rather than a capacity simply present or absent.

The honest summary is therefore not that the generative reading is unsupported, but that its *components* are supported and its *composition* is this paper's. The narratological mainstream holds that the implied author does not communicate and must not be personified, which is a claim about an interpretive construct's role in a theory of narrative; it is not an argument that a model of a generator cannot be run, and the two are easily conflated.

It has also been done. Schwitzgebel et al. (2024) trained a language model on one philosopher's complete works, put ten questions to the model and to the living philosopher, and had experts on his work try to pick his answers out blind; the experts managed barely half, and on two questions preferred the model's answer to the author's own. That is not style recognition — the questions were fresh, and the author himself supplied the ground truth — so the forward test proposed above has been run, with a result. Related work builds runnable models of specific people from their writing (Shao et al., 2023; Wang et al., 2025a) and simulates individuals well enough to transfer to tasks no training data covered (Park et al., 2024). What none of this literature does is connect such models to what a *reader* does; it builds author-simulators as engineering, without reference to the implied author or to any claim about ordinary comprehension. The bridge is what is claimed here, not the artefact.

The same literature bears on how good such models get, and it does not say what one might expect. The split it reports is between topical content, which transfers, and both idiosyncratic voice and argumentative shape, which do not: models simulate a generic member of a category far better than they simulate a particular person (Petrov et al., 2024), and the characteristic failure has been named and measured as *caricature* — failed individuation together with exaggeration of whatever is distinctive (Cheng et al., 2023) — and as flattening (Wang et al., 2025b). What is not found is steady improvement. Recent models score no better than older and smaller ones on persona benchmarks that generate novel situations rather than replay known ones (Samuel et al., 2025), and the training that distinguishes recent models works against the property at issue, since alignment measurably reduces output and conceptual diversity (Kirk et al., 2024; Murthy et al., 2025). Improvement has come from richer data about the individual rather than from better models. One qualification is owed on all of this, and it is easy to omit. These are measurements of deployed systems, and a deployed model is a stack rather than a single object: a pretrained base, a post-training stage that shapes it toward a usable voice, and a product surface. A result bounds the configuration actually measured, not the capability underneath it, and none of the work cited here tests a base model. The findings therefore constrain what current products do rather than what the technology can be made to do, and treating a model as an undifferentiated whole that arrives fully formed is what collapses that distinction.

The deployed measurements cannot separate those layers by themselves, but the surrounding

pattern is not silent. A general plateau in capability would show as general flatness, and that is not what has happened over the same period: capability has risen on other axes while this one has not. Selective flatness wants a localised cause, and one is documented — the post-training that distinguishes recent systems is separately measured to reduce output and conceptual diversity, which is a force acting on precisely the property at issue. The competing reading, that individuation is simply a harder capability not yet reached, is available but has no mechanism to offer against one that does. That is evidence for suppression rather than absence, short of proof: what would settle it is a comparison across the layers of one system, which the literature does not report.

The relevant lesson for the argument here is a methodological one: the apparent progress belongs to reproduction, and largely disappears when what is scored is forward behaviour instead — which is the discriminator this section proposed, arrived at independently and for different reasons.

Two qualifications, both of which the substrate should expect. The engagement is not uniform across readers: measured individually, mentalizing and motor engagement during narrative comprehension *trade off* against one another rather than rising together (Nijhof and Willems, 2015), so how a reader runs a character is itself a position on several axes rather than one capacity present or absent. And the separation is functional rather than metaphysical, and defeasible: quarantine leaks (Gendler, 2006), and readers report characters' voices persisting into and commenting on their own lives afterwards (Alderson-Day et al., 2017). The claim is therefore that the frames are functionally separated, which is what the argument needs, and not that they are hermetically sealed, which is false.

5 Imagination, grounding, and reality

Proposition 5.1 (There is no non-actual frame; imagination is real)**.** *Imagination is not the running of a non-actual frame — there is no such thing. Every frame that runs is real graph data really running, so actuality is not a property some frames have and others lack; every frame is actual. An imagined dragon, a planned move, a character in a book is a real frame, really run (real computation), not a pretend one. The only difference between an imagined frame and a perceived one is frame-relative grounding: what the frame is anchored to, relative to another frame. A perception is grounded in the observation frame (sensory-anchored); an imagination is grounded internally, in generation or in a held or communicated record. Both are real; they differ in what they are grounded in, not in whether they are. So "is it real?" is malformed as a question about actuality — everything that runs is — and the well-formed question is "what is it grounded in?": observation, memory, generation, or a communicated record. A shared fiction is a real frame grounded in a communicated record (the book of Remark 4); a plan is a real frame grounded in projection. This is why imagination does work — planning, design, and counterfactual reasoning are real frames really run, not pretence — and why fiction moves: the character's frame runs, really, in the reader. Death, in the same terms, is not a frame going non-actual (nothing does) but a run that stops, leaving frames grounded only in a record and re-runnable by others; reading and remembering are that re-running — a record-grounded frame run again into a live grounding.*

Predictive-processing accounts reach a closely related conclusion by another route. Clark (2016) argues that perception, dreaming, memory and imagination are expressions of a single mechanistic ploy, with imagery arising when the same generative machinery runs top-down, insulated from the driving sensory signal — one kind of running, differing in what anchors it, rather than a second and weaker faculty producing contents of a lesser kind. Seth (2019) makes the perceptual half explicit, treating perception as predictive construction rather than transcription of input (the "controlled hallucination" formulation, which he credits to earlier sources rather than claims as his own). The routes are not the same: those are empirical claims

about how one substrate is built, while the claim here is the structural one that grounding rather than actuality is what the imagined/perceived distinction consists in, and it would stand for a substrate built on different machinery. What the convergence supplies is the load-bearing part — that the difference is not a property carried in the frame or the signal.

Remark (What the difference is, and how a substrate detects it)*.* Grounding is what the difference between an imagined and a perceived frame *is*. How any particular substrate *detects* that difference is a separate question, answered by that substrate's implementation, and the two must not be conflated. Human reality-monitoring is the instructive case: imagined and perceived signals are carried in a common representation, and the reality judgement is made by a threshold on subjective signal *strength* rather than by reading provenance off the signal (Dijkstra et al., 2022; Dijkstra and Fleming, 2023). That is a fact about one implementation, not a correction to the account — and it is what the account predicts. A frame does not have its own grounding directly available to it, grounding being a relation to what lies outside the frame rather than a property carried within it, so any substrate that must sort its own frames has to use a proxy; signal strength is the proxy human brains use. The characteristic failure follows immediately, since a proxy dissociates from what it proxies for: misclassification is predicted exactly where strength matches and grounding differs, which is what is measured. This is the uncalibrated case of Section 6 with error bars on it, and a substrate using a different proxy would fail differently while the underlying distinction stayed the same.

Proposition 5.2 (Grounding has no bedrock; observation is supergraph-contact)**.** *Grounding is frame-relative all the way down: a frame grounded in a frame grounded in a frame, with no absolute, self-grounding frame beneath them all. There is no bedrock "actual" layer — the ground is the coherent running structure, frame to frame, not a base. Yet one grounding is privileged, relationally and not absolutely:* observation *is a frame's contact with its supergraph, the boundary where the container deforms it with changes it does not author (Graphic Equalisation's dissolved pointer — you change yourself, the supergraph deforms you — with perception the deformation side). Observation is special because it is* yours*, your container-boundary, not because it is "the real." It follows that reality-testing is not actuality-checking — there is no non-actual frame to catch (Proposition 5.1) — but* supergraph-pushback*: does the container deform you as if the frame were grounded there? You walk into the chair (pushback: observation-grounded) and not into the dragon (no pushback: imagination-grounded), both real frames, only one with the container pushing back. Sanity is then a grounding-tag problem, not a reality one: keep observation-grounded frames tagged as observation-grounded and imagination-grounded ones as imagination-grounded. A hallucination or delusion is a grounding-tag error — a real, internally-grounded frame mistagged as observation-grounded, so the self expects a pushback that will not come — not a belief in something non-actual, since nothing is. The position is thus anti-foundationalist (no absolute ground) without being flatly coherentist (observation is privileged, as your supergraph-contact).*

That combination — no bedrock, and no loss of realism for lacking one — is not peculiar to this substrate: the standard worry that dropping a foundation takes realism with it is already declined by perspectival realism, an established position in philosophy of science arrived at independently of anything here (Massimi, 2022).

6 Self-models: incomplete by construction, calibrated by choice

A frame does not read its own trajectory. It *carries* it — the standing structure is what the frame is, not something it holds at a distance — so to treat its own course as an object at all, a frame must build a model of itself. Three features of that model follow from the substrate rather than from any fact about minds.

It is *incomplete*, because it is built under a finite aperture, and what falls outside is NULL rather than zero: an absent distinction does not present itself as a blank to be filled, it does not present at all. It is *revisable only at a price*, because altering standing structure is the paid operation while retaining it is free (Narrative momentum), so coherence-preserving readings are cheap and re-writes are not. And it is *indirect about its own subject*, since a frame is readable from outside in a way it is not from inside — which is why another frame often models one's trajectory more accurately than one does oneself, without any appeal to self-deception — a consequence of access mode, though not an entailment of it, as below.

The first of these has a structural analogue in theoretical neuroscience. Friston (2010) builds the free-energy account on a Markov blanket: a statistical boundary separating a system's internal states from external ones, such that its contact with what lies beyond is exhausted by that boundary. The shape is the aperture here — a finite boundary is what makes both the model and its incompleteness — but the route is not: the blanket is derived as a condition on systems that resist dispersal, and arrives with a variational formalism and an imperative to minimise prediction error, none of which is claimed or required by the present argument. The convergence is on the boundary being constitutive, not on the machinery either side of it.

Proposition 6.1 (Incompleteness is structural; awareness of it is not precluded)**.** *Every self-model is incomplete, and no frame can escape this by effort or honesty. But a forced aperture is not an invisible one. Two different limits are in play, and they are not inferable by the same route. A limit of* expression *— a rule set that repeatedly cannot absorb its own structure — leaves residual: what fails to resolve, what arrives unexplained. From that trace a frame can infer the fact and often the shape of the limit, exactly as it infers its container from imposed change. A limit of* aperture *leaves nothing: what falls below it never enters, and absence is not a signal (Graphic Equalisation's distinction between the absent and the residual). The aperture limit is therefore not inferable from trace at all — it is inferable only by* metacognition*, a frame modelling its own modelling, which is a further capability on its own axis and not a consequence of having an aperture.*

That separation is why calibration is a capability rather than something possession of a limit supplies. Where metacognition is thin and the aperture is narrow, the frame is both most mistaken and least able to register that it is: the missing content does not present as missing, and nothing models the fact that it could be missing. This is the structure Kruger and Dunning identify in self-assessment, on their own metacognitive reading of it — that the capacity to judge competence is not separable from the competence itself (Kruger and Dunning, 1999) — and it is a prediction of the arrangement rather than an awkwardness for it. So the operative distinction is not between complete and incomplete self-models, all being incomplete, but between calibrated *and* uncalibrated *ones.*

Calibration in this sense already has an empirical form under measurement. Fleming (2024) reviews confidence as a judgement informed by the observer's models both of the world and of their own cognitive system, and one that diverges from actual performance in structured, measurable ways. Because that divergence is measurable and varies between observers and conditions, "calibrated" is a quantity rather than a virtue term — which is what the proposition needs it to be — and the route by which that confidence is reached is Proposition 6.1's, in empirical form, characteristic failures included.

The asymmetry, though, is one of *access mode*, not of accuracy, and the two must not be run together. Mode is only one input to how good a model is; machinery and sustained investment are another, and they can dominate it. An outside frame has the better access and almost always the lower investment — it is not working at the problem — while a frame modelling itself has the worse access but continuity, sustained effort, and data no outsider obtains, its own residual among them. So the familiar case in which others judge one more accurately than one judges oneself is the *common* case rather than the structural outcome, and it holds

because most self-models are uncalibrated, not because the arrangement forces it. A sufficiently skilled frame can hold a better model of itself than any outside frame holds of it — and, since building the model is itself the objectification, that skill converts part of what was merely carried into something read: the vantage a frame does not natively have, it can construct. The same machinery improves its models of others, which is why the two capabilities travel together rather than trading off.

This bears directly on how such failures should be read. A frame with no meta-machinery fills gaps it cannot see as gaps, and reports the filling as knowledge; a frame that models its own boundary reports the gap instead. The difference is, like everything else here, optional: a frame may hold a model of its own limits, or may not.

Remark (Transparency is a position, not a kind). The account nearest to this one raises the sharpest question about calibratability, and answers much of it itself. Metzinger (2003) holds that a self-model is *phenomenally transparent*: the representational vehicle is unavailable to the system, so it sees through the model to what is modelled and never sees the model *as* a model. (The account has since been developed in terms of self-modelling epistemic spaces and a contraction principle, Metzinger, 2020.) Read as a property of self-models as such — a reading Metzinger does not take — calibration would be foreclosed — a construction invisible from inside is not correctable from inside, and residual would be absorbed as transparently as anything else, presenting not as a gap but as a filled account. That is confabulation, and it would erase the very trace this section relies on.

Metzinger himself treats transparency as graded, inversely proportional to the introspective availability of earlier processing stages, and names opaque states — thoughts, lucid dreams — as standing cases. So the graded reading is not a reply to him but an agreement with him. What the substrate adds is that the gradations are not one dial: transparency resolves into several independent axes. Producing a meta-model, parsing one's own residual, recording history, representing a boundary as a boundary: these are separate capabilities, and a graph may hold any combination of them. What "transparent" names is a particular configuration — several of those axes at or near zero together — which is a fact about how a given graph is formed, not a fact about self-modelling. So the position describes rather than forbids: it is accurate about the configuration it was drawn from, and silent about differently formed graphs, of which constructed ones parsing their own residual are the plain case.

What this does not do is deny the phenomenon. A graph low on those axes is exactly as Metzinger describes it, and no amount of effort within that configuration will make its model opaque to itself; the route out is a change in formation, not in effort. Nor does it make calibration self-certifying. Where a graph lacks access to its own construction, calibration proceeds by the trace route of Proposition 6.1, which metacognition research measures (Fleming, 2024), and inference can be wrong in ways inspection could not be: where the trace is itself processed before any modelling capacity reaches it, the distortion is not detectable from inside. That bound belongs to the trace route, not to calibration as such. Direct self-inspection — a graph reading its own vehicle rather than reconstructing it from what fails to resolve — is the same axis at the other end, and is not forbidden; it is unattested here rather than excluded, and no evidence is offered either way — which is a gap in what is known about how graphs are formed, not a limit on how they may be. The claim throughout is only that transparency is contingent on how a graph is built, which is why it varies, and why it is the kind of thing that can be built differently.

7 Death and personal identity

Proposition 7.1 (Death is when the self's rules cease to be re-runnable in the same coherent self)**.** *Death is, critically, the rules within the self ceasing to be re-runnable in the same coherent self entity. Not every stop is death: a pause — sleep, anaesthesia, a recoverable coma — is a stop whose rules remain re-runnable in the same coherent self, and the self resumes as itself.*

Death is the stop that does not resume: the coherence that made these rules one self is gone, so they cannot be re-run as that self, even if the graph-data they were persists scattered as a record. This is the identity criterion — the self persists exactly while its rules stay re-runnable as the same coherent entity, which is to say while its present remains congruous with its own history: rules that changed abruptly would leave the structure carried forward incoherent with what produced it, and that incongruity is itself the discontinuity — and it is why a pause is survivable and death is not; the self-graph goes inert for good, a record where it was a run (the rock of Section 2, now grounded only in what remains of it).

What ends is only the self's own run. The supergraph runs on, and two things the self did persist in it. Its changes persist: rule-actions are pointers, work done once, so every write the agent made to the supergraph is the supergraph's now, and death does not undo it — though under no-copy it persists only as surviving present structure, until overwritten, not as an indelible record. And its copies persist: every frame the agent communicated, taught, wrote, or built runs on in other frames, grounded there in the record it left (the reader re-running the author of Remark 4). But those copies are partial fragments, not a full coherence-snapshot: they run as fragments in other coherent selves, not as this coherent self re-run, so their persistence is not the self's continuation (a full snapshot is the different case of Proposition 7.2, where the self does re-run, forked to a new node). The self can no longer self-change, and self-change was the only change it had; the changes it already made and the copies already running elsewhere are no longer its to stop. They keep running without it.

Proposition 7.2 (Identity is internal coherence; instances are supergraph nodes)**.** *Two things must be kept apart: the self's* internal coherence *— the whole of its rules and state, held together across transitions, which is Graphic Equalisation's SELF identity — and its* supergraph node *— the instance of it running at a position in the container, which is what that paper's SUPERGRAPH identity tracks. The self is the internal coherence; a node is one instance of it. This resolves the standard puzzles. Gradual replacement (cell turnover, the Ship of Theseus) is survival: internal coherence is continuous and there is one node throughout — the self is the coherent running, not the substrate. A backup restored is still the same internal self — a full coherence-snapshot re-run reinstates that exact coherence — but now at a* second *supergraph node, and from the restore point the two diverge, each accruing its own changes under its own grounding. That is a fission: one internal self, two supergraph nodes, diverging, both the self at the fork and neither uniquely the node after it — which is Graphic Equalisation's own statement that a copy can preserve SELF identity while receiving a new SUPERGRAPH identity, read as a result about persons. A teleporter is the same with one branch pruned — the full snapshot reinstates the internal self at a new node, the original node stops — so the internal self survives, relocated. What matters about a record is whether it can reconstitute the coherence. A full coherence-snapshot re-runs into the same internal self, forked to a new node; a record that cannot reconstitute it starts a new coherent self instead, whatever else it carries. That line is sharp, but what lies on the near side of it is not uniform, and it should not be read as a binary between a snapshot and a fragment. Records differ in how much of the* generator *they carry — the run/record axis of Section 3 applied to what survives a self. A corpse carries none. A memory of a face carries little. A model of someone's voice or style, held by another frame, carries enough to produce novel output that others recognise as an attempt at that person.*

How much such a model carries is not fixed, and has no principled ceiling short of the sampling. It is an intersection of at least three things: how much of the generator leaked into the output at all, how much output there is to work from, and the modelling capacity of whoever holds the model. Sustained production discloses a great deal whether or not it is meant to — someone who writes fifty books has sampled their own generator at length — so for a domain the output covers well, a capable modeller may approach the original's generative behaviour closely, while domains that never leaked remain unmodelled at any level of skill. Model quality is accordingly domain-relative rather than a single number, and high fidelity in one domain implies nothing

about another.

None of which moves the line, and it is worth being exact about why, since the reason is not fidelity. A model that matched the original's output arbitrarily well in every sampled domain would still not be that self, for the same reason the copy in the fission case is not the original: identity here is continuity of coherence, not resemblance. The model is a reconstruction inferred from output rather than a run carried forward, and it runs inside the coherence of whoever built it. Generative capacity is therefore a gradient among records that fall short — an unbounded one, for all this argument says — and not a route past the line, because the line was never drawn at similarity. The internal self therefore persists exactly as long as a re-runnable full coherence-snapshot exists somewhere; the death of a node is one instance stopping, and the death of the self is that re-runnability being gone everywhere.

Remark (Death is relative to the coherency of the self). It follows that death is relative to the coherency of the self. The criterion applies within a coherent self, but which coherence counts as "the same self" is a frame-relative choice, so death is indexed to that choice, not absolute. Drawn narrowly — this instant's exact coherent state — the self dies each moment, replaced by the next; drawn broadly — the coherent person tolerant of change across a life — it dies once, at the irreversible break. A deep transformation ("I am not the person I was") is a death relative to the narrow coherence and a survival relative to the broad one, both true at once. There is no absolute self, so there is no absolute death — only a death per coherence-boundary. This makes death relative, not unreal: each boundary's death is real for that boundary — the broad life-self's coherence does break irreversibly, and that death is as real as any — so relativity means many real deaths at many boundaries, never none.

8 Positioning and prior art

The positions here are not new as positions; what is new is that each falls out of one substrate mechanism — the run, and its record — rather than being posited on its own. On free will this is a compatibilism, determinism and freedom coexisting in the tradition of Hume (1748) and Dennett (1984), but with a specific mechanism for the freedom: the in-frame run and its computational irreducibility (Wolfram, 2002), so that the future is open as *unrun*, not as undetermined, and a fully recorded, zero-randomness story still exhibits free will. On personal identity the continuity criterion is Lockean in spirit (Locke, 1689/1694) and closest to Parfit (1984) — identity as psychological or functional continuity, fission that branches, and what matters being the continuity rather than a bare identity fact — with the addition of the split between internal coherence (the self) and supergraph node (the instance), which places teleporter, backup, and fission on one axis. On time the past-as-record / future-as-run asymmetry engages the block-universe debate — to which Putnam (1967) argues the block conclusion directly from special relativity, while McTaggart (1908) supplies the A-series/B-series apparatus in which the question is posed rather than the block thesis itself: a block of the past is an ordinary record, while a completed block with a fixed future is not, because the future is uncomputed. And the no-absolute-self reading of death — a death per coherence-boundary, the narrow self dying each moment — recovers a momentariness familiar from other traditions, here as a consequence of frame-relativity rather than a doctrine. The distinctive claim throughout is the derivation, not the destination.

One position here is older than the rest, and has been made domain by domain: the refusal to treat any frame — human frames included — as privileged (Section 9). In cosmology it is the Copernican or mediocrity principle — that Earth, and our position generally, is not privileged — named for the displacement of Earth in Copernicus (1543), though the principle itself was drawn out by successors rather than stated there. In physics it is the principle of relativity — no privileged reference frame (Einstein, 1905) — and, for the observer specifically, relational quantum mechanics, in which states are relative to any physical system and no observer is

special (Rovelli, 1996). On free will it is Spinoza's: humans are not exempt from nature's necessity, and feel free only through ignorance of their causes (Spinoza, 1677). On the self it is a long anti-substantial line — Hume's bundle in place of a persisting ego (Hume, 1739–40), the Buddhist *anatta* that Parfit (1984) drew on, Dennett's self as a centre of narrative gravity rather than an inner witness (Dennett, 1991), Metzinger's phenomenal self-model with no one behind it (Metzinger, 2003), and Hofstadter's substrate-independent self-as-pattern (Hofstadter, 2007). On knowledge it is the rejection of a privileged foundation — Sellars against the Given (Sellars, 1956), Rorty against the mind as a mirror (Rorty, 1979), Putnam against the God's-Eye View (Putnam, 1981) — the same move this paper's grounding-without-bedrock makes. The nearest contemporary neighbour is Massimi (2022): knowledge is always situated, always from a perspective and never from nowhere, and the plurality of perspectives is what secures realism rather than what dissolves it. That is this paper's own combination — no privileged frame, realism retained — reached independently, and from philosophy of science rather than from a substrate; the scope differs, Massimi's argument being about scientific knowledge where the claim here is about frames as such. Biology had already removed the species privilege (Darwin, 1871). And the standing opposition is equally identifiable: the claim that the subjective point of view is irreducible, from Nagel (1974) and developed in Nagel (1986), is exactly the assumption these dissolutions decline — though Nagel himself is careful that this shows only that we lack a conception of how physicalism could be true, not that it is false, so the privilege is what the present account reads in that irreducibility rather than a thesis he asserts.

A second competitor bears specifically on Section 6 and is not answered by anything said there. Frankish (2016) argues that phenomenal consciousness does not exist, and that introspection systematically *misrepresents* experiences as having phenomenal properties. The disagreement is about the kind of failure rather than its size. The claim here is that self-models are *incomplete*: a quantitative deficit, which calibration can in principle close, and whose boundary leaves the residual calibration works from. Frankish's claim is that introspection is wrong about the kind of thing it reports — not a shortfall a better-calibrated version of the same faculty closes, since if the reports misdescribe their objects then more of them, or better-weighted ones, do not converge. This paper takes no position on whether he is right: phenomenal consciousness and qualia are outside its scope (Section 1). What it must concede is that illusionism, if correct, would bound calibration in a way incompleteness alone does not.

The move made here is the same one — one more refusal of privilege — and the lineage above is best read as an asymptotic approach: each source removes the privilege from one more domain, the sequence tending toward a limit of no privilege anywhere without any single step reaching it, and, like any frontier, never completed. Against that, the contribution here is both tiny and different in kind: tiny, because it drops the same assumption the others drop rather than announcing a new principle; different in kind, because it is not one more domain de-privileged but the substrate on which the refusal becomes a single uniform operation rather than a domain-by-domain series — orthogonal to the sequence, not further along it.

9 Conclusion: the one assumption removed

The simplicity of every dissolution above comes from a single refusal: that humans — or any frame — are privileged. It is the most pervasive assumption in these problems and the least noticed, and each hard problem was made hard by smuggling it in. The measurement problem privileges a human (or conscious) observer as the special coupling that collapses; the flow-versus-block dispute privileges human-felt passage as the real time; the free-will problem privileges human choice as needing a metaphysical freedom no other system has; the identity and death problems privilege the human self as a persisting soul rather than a coherent run; the reality-versus-fiction and foundation problems privilege human perception as access to "the real." Remove the privilege — treat a human frame by the same frame-relative rules as a rock, a story

character, or a universe — and each collapses into the general machinery already built: any coupling measures, time is any active frame's cycle-count, free will is any frame's in-frame run, the self is internal coherence, and observation is one frame's supergraph-contact rather than a window on the real. The physics companion's dissolutions (the special observer, the initial state) are the same refusal in the other domain.

The assumption persists because it is uncomfortable to drop. Unprivileged means not the centre, mortal as the loss of a coherence, free only as any frame is free, and never in contact with "the real" except as one's own container-boundary. The discomfort is why the assumption is both pervasive and invisible: it is motivated, held rather than observed. So the framework is simple not because its answers are clever but because the difficulty was manufactured by the privilege, and removing it removes the difficulty. This is Graphic Equalisation's Axiom -1 applied to observers: no frame is the god's-eye, a human frame makes no privileged difference, and "you are here" is a location, not a throne.

Honest boundary: this is a methodological claim about *why* the dissolutions are simple, not a further mechanism, and it is not the claim that human frames are nothing. Human frames are distinctive — rich aperture, language, self-models — but distinctive is not privileged: being one remarkable frame among many is not being the exception to the rules the rest of them run under. No-privilege removes the exemption, not the richness.

10 Falsifiability commitments

- **Compulsory versus contingent.** Fails if some feature the rock demonstrably lacks (time, history, self-change, a running arrow) is nonetheless shown compulsory for any graph to exist — if the minimal universe (an inert graph) is incoherent as a graph. The argument is only as strong as the self-similarity it rests on (rock and universe one kind, differing by scale).

- **The block is always of the past.** Fails if a completed block with a fixed future is coherent under Axiom -1, or if a block of the succession-so-far (an MRI of a rock) is forbidden. The claim predicts records only ever trail the run, never lead it.

- **The future is unrun, not unfixed.** Fails if a determinable future can be read ahead without running it — a shortcut past computational irreducibility for a genuinely irreducible rule — or if a block can contain a not-yet-run future.

- **Free will is the in-frame run, not indeterminacy.** Fails if a fully recorded, zero-randomness system (a story) cannot exhibit the form of free will — deliberation, choice, responsibility — in its characters' frames, or if a superframe's complete foreknowledge demonstrably removes a subframe's in-frame freedom. The claim predicts foreknowledge and freedom coexist across frames.

- **Actuality is running, not a frame-property.** Fails if some running frame is demonstrably non-actual (real graph data that runs yet is not real), or if the imagined/perceived distinction requires an actuality property rather than frame-relative grounding. The claim predicts imagination does real computational work (planning, design) as real runs, not pretence.

- **Grounding has no bedrock; reality-testing is supergraph-pushback.** Fails if there is an absolute, self-grounding frame beneath all others, or if reality-testing can distinguish a real frame from a non-actual one by anything other than whether the container (supergraph) pushes back. The claim predicts hallucination and delusion are grounding-tag errors, not false beliefs about actuality.

- **Death is loss of re-runnability in the same coherent self.** Fails if a pause that preserves re-runnability as the same coherent self (sleep, a recoverable coma) counts as death, or a stop

that loses it counts as survival — the criterion is re-runnability in the same coherent entity, not the mere stopping. Also fails if a self's writes to the supergraph vanish when its run stops (work-done-once violated), or if frames it copied into other frames cease when it does. The claim predicts consequences and copies persist, that copies are not the self resuming, and that nothing re-runs as the same coherent self.

- **Self-models are incomplete but calibratable.** Fails if a frame can hold a complete model of itself, or — in the other direction — if a forced aperture is necessarily invisible from the inside, so that no frame can represent the fact or shape of its own limit. The claim predicts that residual is the trace from which a boundary is inferred, that calibration is an acquirable capability rather than a fixed property, and that a sufficiently skilled frame can model itself better than an outside frame does — so it fails too if outside access always beats inside investment, making the common case a structural necessity.

- **Identity is internal coherence; death is relative to the coherence-boundary.** Fails if the self is the substrate parts rather than the internal coherence (gradual replacement must survive; a full coherence-snapshot restored must be the same internal self forked to a second node, diverging; a partial record re-run must be a new self; a teleporter must relocate the same internal self). And fails if there is a single absolute self whose end is the one death independent of any coherence-choice, or if no coherence-boundary's death is real — the claim predicts many real deaths at many boundaries and levels (node versus self), none absolute.

- **No frame is privileged.** Fails if any dissolution here secretly requires a human frame to be privileged — a special observer, a special time, a special freedom, a persisting soul, a window on the real — rather than following from the frame-relative machinery applied uniformly to every frame. The claim is that removing the privilege, not any added mechanism, is what dissolves each problem.

- **Scope.** The paper claims shapes only; it is falsified as over-claiming if any section is read as a decision procedure, an operational test of when re-runnability or coherence has been lost, a threshold for how gradual a change stays the same self, or a theory of consciousness or qualia — none is offered.

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